

# Technical Learning Report: Day 1

**Date:** Sep 1, 2026

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**Subject:** Week 1, Day 1 — Testing Fundamentals

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## 1. Objective

The objective of Day 1 was to build a foundational understanding of why software testing exists as a discipline. This involved understanding the relationship between human error, defects, and system failure, distinguishing testing from debugging and from quality assurance, and studying the seven ISTQB testing principles. The notes reflect a focused study of these core concepts along with worked examples connecting the seven principles to real-world testing scenarios.

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## 2. Technical Breakdown

### A. Software Testing: Definition and Purpose

- Software testing is a set of steps taken to discover defects and evaluate the quality of a working product.
- It helps maintain a standard level of quality code, reducing the risk of system failure during operation.
- Testing is necessary because it ensures that the quality of the project matches its objectives, while staying within the set scope, time, and budget.

### B. Testing vs. Debugging

- Testing is the process of checking whether the software works as expected. Its aim is to identify defects — the guiding question is "Is there any problem?"
- Debugging focuses on fixing the cause of defects. Its aim is to remove bugs — the guiding question is "Why is there a problem, and how do we fix it?"

### C. Testing vs. Quality Assurance (QA)

- Testing checks the product to find existing problems.
- QA improves the overall process so that fewer problems occur in the first place.

#### D. Error, Defect, Failure, and Root Cause

- An error is a mistake made by a person.
- That error can lead to the creation of a defect.
- When the system fails due to a defect, this is considered a failure.
- The main reason behind the problem is referred to as the root cause.

#### E. The Seven Testing Principles

1. Testing shows the presence of defects, not their absence.
  2. Exhaustive testing (testing every single thing) is impossible.
  3. Early testing saves more time and money.
  4. Defects tend to cluster together.
  5. Repeated use of the same tests loses effectiveness over time (the "pesticide paradox") — tests must be changed or new ones added.
  6. Testing is context-dependent — there is no single universal way of testing.
  7. Absence of bugs/defects is a fallacy — a defect-free system does not necessarily mean it meets the required criteria.
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### 3. Practical Implementation

The practical work on Day 1 focused on applying the testing principles through examples:

1. **Principle 1 – Testing shows defects:** No defects were found during testing, but users later could not access the website due to external issues, showing that testing cannot guarantee a defect-free system.
2. **Principle 3 – Early testing saves money:** Poor testing before release caused production failures, resulting in loss of time, money, and reputation.
3. **Principle 5 – Pesticide Paradox:** Using the same old test cases repeatedly may miss defects in new features, so test cases should be updated.
4. **Principle 6 – Context-dependent testing:** Testing methods should be selected based on the type and context of the software.

5. **Exhaustive testing is impossible:** Testing every module and possible scenario is impractical, so testers must prioritize important areas.
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## 4. Key Findings & Best Practices

- Completing testing with no defects found does not guarantee the absence of real-world failures — other factors (e.g., infrastructure or ISP-related issues) can still cause a system to fail.
  - Pushing code to production without proper testing can lead to tangible business costs: loss of money, time, or reputation.
  - Relying on outdated test scripts without updating them for new functions or features undermines the value of testing.
  - There is no universal, fixed method for checking code — testers and QA must adapt their approach to the specific system being tested.
  - Testing every single module is not feasible; over-scripting for every module can result in more test code than actual production code.
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## 5. Conclusion

Day 1 established the conceptual foundation of software testing by clarifying the distinctions between testing, debugging, and quality assurance, and by tracing the chain from human error to defect to failure. The seven ISTQB testing principles were studied and reinforced through five worked examples connecting theory to practical scenarios. This groundwork in core testing terminology and principles provides the conceptual basis for the case study, tooling, and AI-assisted exercises expected later in the Day 1 assignment scope.

Day I.

Software Testing is a set of steps taken to discover defects, evaluate the quality of working product. It ~~make~~ maintain a standered level of quality code. helping reducing the risk of sys failure during the operation.

Why Testing Necessary?

It ensure that the quality of the project is meatched with all the objective where meet within the set scope, time and budget.

Testing and Debugging

Testing is the process of checking whether the software works as expected.

Aim to identify defect. "Is there any problem".

Debugging focuses on Fixing the cause of defects. It's aim is to remove the bugs. "Why is there a problem, and how do we fix it".

## Testing and Quality Assurance (QA)

Testing checks the product to find problems. QA improves the process so few problems occur in the first place.

## Error, Defects, Failure, and Root Causes

- An error is a mistake made by a person
- That error can lead to creating a defect
- When the system failed due to the defect
- Main reason behind the problem root cause.

## Testing Principle

1. Testing shows the presence, not the absence of defects.
2. Testing every single thing is impossible
3. Early testing save more money
4. Cluster the defects together
5. Change the tests or add new tests.
6. There is not a universal way of testing
7. Absence of bug/defect does not mean the system meet the criteria.

### Examples:-

1. Testing is completed and no defects were initially found, code is pushed to the production. but when real user try to login or visit the site there is no service responding due to isp or other reason we have done ever thing from our side but ~~at~~ there can be multiple reason it main fail "Tester might miss the part of code that is creating a failure". (1st principle)

2. Company push code without doing proper testing and it cause failure in production leading to loss of money, time or business reputation. (3rd principle)

3. Tester and QA uses outdated test scripts not writing new script with the introduction with new function or features. (5th principle)

4. There is no fix way of checking the code the tester and QA need to find appropriate ways to check the code. (6th principle)

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5. Testing every single module is not feasible so tester are not specifically required to make scripts for each module they might change ~~can~~ constantly. ~~and~~ If we try to make it there will be more test cases than the actual code.